Fermion path integral

$$Z = \int \langle -\bar{\psi}_0 | 1-\varepsilon H | \psi_{N-1} \rangle e^{-\bar{\psi}_{N-1}\psi_{N-1}} \langle \bar{\psi}_{N-1} | 1-\varepsilon H | \psi_{N-2} \rangle e^{-\bar{\psi}_{N-2}\psi_{N-2}}$$

$$\ldots \langle \bar{\psi}_1 | 1-\varepsilon H | \psi_0 \rangle e^{-\bar{\psi}_0 \psi_0} \prod_{i=0}^{N-1} d\bar{\psi}_i \, d\psi_i$$

For some j: $\langle \bar{\psi}_{j+1} | 1-\varepsilon H(\hat{\Psi}^+, \hat{\Psi}) | \psi_j \rangle =$ (by construction, $\hat{\Psi}|\psi_j\rangle = \psi_j|\psi_j\rangle$, $\langle\bar{\psi}_{j+1}|\hat{\Psi}^+ = \langle\bar{\psi}_{j+1}|\bar{\psi}_{j+1}$)

$$= \langle \bar{\psi}_{j+1} | \psi_j \rangle - \varepsilon \langle \bar{\psi}_{j+1} | \psi_j \rangle H(\bar{\psi}_{j+1}, \psi_j) =$$

$$= \langle \bar{\psi}_{j+1} | \psi_j \rangle \left(1 - \varepsilon H(\bar{\psi}_{j+1}, \psi_j)\right) =$$

$$= e^{\bar{\psi}_{j+1}\psi_j} e^{-\varepsilon H(\bar{\psi}_{j+1}, \psi_j)}$$

Next, we introduce $\bar{\psi}_N = -\bar{\psi}_0$, $\psi_N = -\psi_0$ (antiperiodic boundary conditions!) ABC so that we can write

$$Z = \int \prod_{i=0}^{N-1} e^{\bar{\psi}_{i+1}\psi_i} e^{-\varepsilon H(\bar{\psi}_{i+1}, \psi_i)} e^{-\bar{\psi}_i \psi_i} d\bar{\psi}_i \, d\psi_i =$$

$$= \int \prod_{i=0}^{N-1} \exp\left\{\left(\frac{\bar{\psi}_{i+1} - \bar{\psi}_i}{\varepsilon} \psi_i - H(\bar{\psi}_{i+1}, \psi_i)\right)\varepsilon\right\} d\bar{\psi}_i \, d\psi_i$$

Here $\dfrac{\bar{\psi}_{i+1} - \bar{\psi}_i}{\varepsilon} \psi_i \cong \partial_\tau \bar{\psi} \psi = -\bar{\psi} \partial_\tau \psi$

$$Z \Rightarrow \int \exp\left\{\int_0^\beta d\tau \, \bar{\psi}(\tau)\left(-\frac{\partial}{\partial\tau} - \Omega_0 + \mu\right)\psi(\tau)\right\} [\mathcal{D}\bar{\psi}\mathcal{D}\psi].$$

where the meaning of $\frac{1}{\varepsilon}(\bar{\psi}_{i+1} - \bar{\psi}_i) \to \partial_\tau \bar{\psi}$ replacement is discussed in p.648 of Shankar's book (Ch.21).

Observe now that ABC: $\psi_N = \psi(\varepsilon N) = \psi(\beta) = -\psi_0 = -\psi(0)$

$$\bar{\psi}_N = \bar{\psi}(\beta) = -\bar{\psi}(0)$$

Expand in Fourier modes $\psi(\tau) = \frac{1}{\sqrt{\beta}} \sum_n e^{-i\omega_n \tau} \psi(\omega)$

$$\bar{\psi}(\tau) = \frac{1}{\sqrt{\beta}} \sum_n e^{i\omega_n \tau} \bar{\psi}(\omega)$$

$$\psi(\tau=\beta) = -\psi(\tau=0)$$

$$\Rightarrow \quad \frac{1}{\sqrt{\beta}} \sum_n e^{-i\omega_n \beta} \psi(\omega) = -\frac{1}{\sqrt{\beta}} \sum_n e^{-i\omega_n 0} \psi(\omega)$$

$$\Rightarrow \quad e^{-i\omega_n \beta} = -1 \text{ for every } \underline{n}$$

$$\Rightarrow \quad \omega_n \beta = (2n+1)\pi$$

$$\omega_n = \frac{(2n+1)\pi}{\beta} = 2\pi\left(n+\tfrac{1}{2}\right) T$$ fermion Matsubara frequency.

$T \to 0$ $(\beta \to \infty)$ limit: $\frac{1}{\beta} \sum_n \ldots = \int \frac{d\omega}{2\pi} \ldots$

$$\int_0^\beta d\tau\, \bar{\psi}(\tau) (-\partial_\tau) \psi(\tau) = \frac{1}{\beta} \sum_{n,m} \int_0^\beta d\tau\, e^{i\omega_n \tau} \left(-\frac{\partial}{\partial \tau}\right) e^{-i\omega_m \tau} \bar{\psi}(\omega_n) \cdot \psi(\omega_m)$$

$$= \frac{1}{\beta} \sum_{nm} \underbrace{\int_0^\beta d\tau\, e^{i(\omega_n - \omega_m)\tau}}\, i\omega_m \bar{\psi}(\omega_n) \psi(\omega_m) = \sum_n i\omega_n \bar{\psi}(\omega_n) \cdot \psi(\omega_n)$$

$$\tau = \beta x, \quad \omega_n - \omega_m = 2\pi(n-m)\beta^{-1}$$

$$\beta \int_0^1 dx\, e^{i\frac{2\pi}{\beta}(n-m)\beta x} = \beta \int_0^1 dx\, e^{i 2\pi (n-m) x} = \beta \delta_{nm}$$

$$Z = \int \left[\mathcal{D}\bar{\psi}(\omega) \mathcal{D}\psi(\omega)\right] \exp\left\{\sum_n \bar{\psi}(\omega_n)(i\omega_n - \Omega_0 + \mu)\psi(\omega_n)\right\}.$$

Consider $\langle \bar{\Psi}(\omega_1)\,\Psi(\omega_2)\rangle = \dfrac{\int \bar{\Psi}(\omega_1)\,\Psi(\omega_2)\, e^{\sum_n \bar{\Psi}(\omega_n)(i\omega_n - \Omega_0+\mu)\Psi(\omega_n)}\, D\bar{\Psi}D\Psi}{\int e^{\sum_n \bar{\Psi}(\omega_n)(i\omega_n - \Omega_0+\mu)\Psi(\omega_n)}\, D\bar{\Psi}D\Psi}$

Clearly must be diagonal in $\omega \Rightarrow \omega_1 = \omega_2 = \omega_k$ (for some particular k)

$$\langle \bar{\Psi}(\omega_k)\,\Psi(\omega_k)\rangle = \frac{\int \bar{\Psi}(\omega_k)\,\Psi(\omega_k)\, e^{\bar{\Psi}(\omega_k)(i\omega_k - \Omega_0+\mu)\Psi(\omega_k)}\, d\bar{\Psi}_k d\Psi_k}{\int e^{\bar{\Psi}(\omega_k)(i\omega_k - \Omega_0+\mu)\Psi(\omega_k)}\, d\bar{\Psi}_k d\Psi_k}$$

$$= \frac{+1}{i\omega_k - \Omega_0+\mu}$$

since $\int \bar{\Psi}\Psi\,(1 - A\bar{\Psi}\Psi)\, d\bar{\Psi}d\Psi = -1.$

$\int (1 - \bar{A}\bar{\Psi}\Psi)\, d\bar{\Psi}d\Psi = A.$

Then $\langle \bar{\Psi}(\tau_1)\,\Psi(\tau_2)\rangle = \dfrac{1}{\beta}\sum_{n,m} e^{i\omega_n\tau_1 - i\omega_m\tau_2}\, \underbrace{\langle \bar{\Psi}(\omega_n)\,\Psi(\omega_m)\rangle}_{\frac{+\delta_{nm}}{i\omega_n - \Omega_0+\mu}} =$

$$= \frac{1}{\beta}\sum_n e^{i\omega_n(\tau_1-\tau_2)}\,\frac{+1}{i\omega_n - \Omega_0+\mu}$$

Consider $T\to 0$ limit of this object:

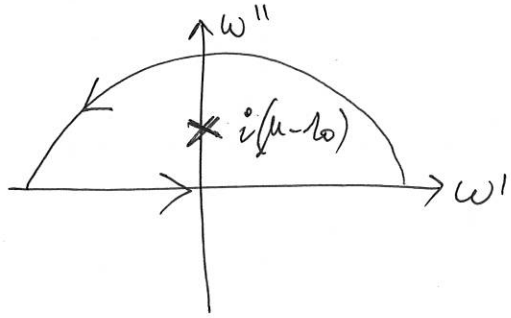
$$\langle \bar{\Psi}(\tau_1)\,\Psi(\tau_2)\rangle = \int \frac{d\omega}{2\pi}\,\frac{+e^{i\omega(\tau_1-\tau_2)}}{i\omega - \Omega_0+\mu} = +\begin{cases} \theta(\mu-\Omega_0)\, e^{-(\mu-\Omega_0)\tau}, & \tau = \tau_1-\tau_2 > 0; \\ -\theta(\Omega_0-\mu)\, e^{-(\mu-\Omega_0)\tau} & \tau < 0 \end{cases}$$

Thus, $\tau \to +0$ limit $(\tau_1 = \tau_2 + 0)$

$\langle \bar{\Psi}(\tau_1)\,\Psi(\tau_2)\rangle = \theta(\mu-\Omega_0)$ = Fermi-Dirac function at $T=0$

Suppose $\tau = \tau_1 - \tau_2 > 0$:

$$n(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{i\omega\tau}}{i\omega - \Omega_0 + \mu} = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi i} \frac{e^{i\omega\tau}}{\omega + i(\Omega_0 - \mu)} = \frac{2\pi i}{2\pi i}\, \theta(\mu - \Omega_0) \cdot e^{i\tau i(\mu - \Omega_0)}$$



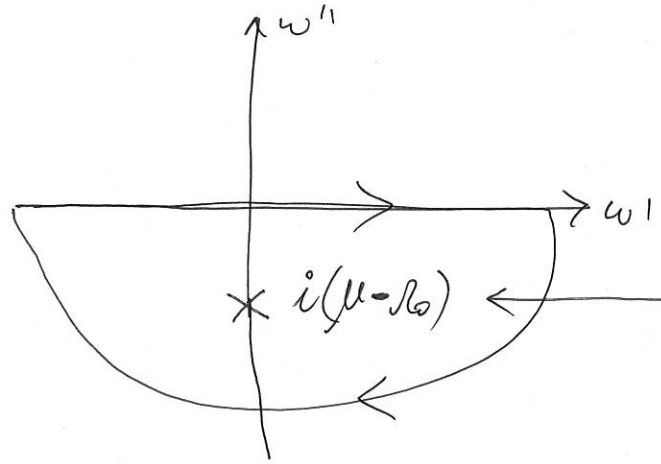
$$e^{i\omega\tau} = e^{i\tau(\omega' + i\omega'')} = e^{i\omega'\tau} \cdot e^{-\omega''\tau}$$

converges in $\underline{\omega'' > 0}$ half-plane

$$i\omega = \Omega_0 - \mu \quad , \omega = -i(\Omega_0 - \mu) = i(\mu - \Omega_0)$$

$\Rightarrow$ for $\tau > 0$: $n(\tau) = \theta(\mu - \Omega_0)\, e^{-(\mu - \Omega_0)\tau}$

Now let $\tau < 0 \Rightarrow e^{i\omega\tau} = e^{i\omega'\tau} e^{-\omega''\tau} \Rightarrow$ need $\omega'' < 0$



for $\mu - \Omega_0 < 0$

$$n(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi i} \frac{e^{i\omega\tau}}{\omega - i(\mu - \Omega_0)} = \frac{-2\pi i}{2\pi i}\, e^{i\tau(\mu - \Omega_0)i}\, \theta(\Omega_0 - \mu)$$

$$= -\,\theta(\Omega_0 - \mu)\, e^{+\tau(\Omega_0 - \mu)}$$

Free fermions $\hat{H}_0 = \sum_{\vec{k}} \left(\frac{\hbar^2 \vec{k}^2}{2m} - \mu \right) \underbrace{\hat{\psi}^+_k \hat{\psi}_k}_{\text{operators}}$

$$\Rightarrow \quad Z = \int D\bar{\psi} D\psi \; \exp \sum_{\vec{k}, \omega_n} \left(i\omega_n - \frac{\hbar^2 \vec{k}^2}{2m} + \mu \right) \underbrace{\bar{\psi}_k(\omega_n) \psi_k(\omega_n)}_{\text{Grassmann numbers}}$$

Interacting fermions $\hat{H}_0 \to \hat{H}_0 + \hat{H}_{int}$

<u>Superconductivity</u> (reduced BCS Hamiltonian)

$$\hat{H}_{int} = -\frac{g}{V} \sum_{\vec{k}, \vec{k}'} \hat{\psi}^+_{k\uparrow} \hat{\psi}^+_{-k\downarrow} \hat{\psi}_{-k'\downarrow} \hat{\psi}_{k'\uparrow}$$

spin indices $\uparrow, \downarrow$